Exact Line Search

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Ex $f(x,y) = 4x^2 - 4xy + 2y^2$

Apply ~~steepest descent~~ by exact line search using $x_0 = (2,3)$ initial guess

$$\boxed{x_{k+1} = x_k + \alpha_k p_k}$$

α_k: step length; p_k: descent direction

$$\boxed{p_k = -\nabla f(x_k)}$$

exact line search

$k=0$

$x_1 = x_0 + \alpha_0 p_0$ $\quad p_0 = -\nabla f(x_0)$

$x_1 = (2,3) + \alpha_0 p_0$

$x_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \alpha_0 \begin{bmatrix} -4 \\ -4 \end{bmatrix}$

$\rightarrow x_1 = \begin{bmatrix} 2 - 4\alpha_0 \\ 3 - 4\alpha_0 \end{bmatrix} \begin{matrix} \rightarrow x \\ \rightarrow y \end{matrix}$

$\nabla f(x) = \begin{bmatrix} 8x - 4y \\ -4x + 4y \end{bmatrix}$

$\nabla f(x_0) = \begin{bmatrix} 16 - 12 \\ -8 + 12 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ $\quad (2,3)$

$\min f(x_1)$

$\min f(x_1) = \underbrace{4(2+4\alpha_0)^2 - 4(2+4\alpha_0)(3+4\alpha_0) + 2(3+4\alpha_0)^2}_{\phi(\alpha_0)}$

$\min_{\alpha_0} \phi(\alpha_0) \rightarrow$ 1 variable calculus.

$\phi'(\alpha_0) = 0 \Rightarrow 4 \cdot 2 \cdot 4(2+4\alpha_0) - - - - - - = 0$

$\alpha_0 = -1/2$ ✓ critical pt.

$\phi''(1/2) > 0$ ✓ (2nd deriv. test)?

$x_1 = \begin{bmatrix} 2 - \cancel{4} \cdot \frac{1}{\cancel{2}} \\ 3 - \cancel{4} \cdot \frac{1}{\cancel{2}} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\nabla f(x_1) \overset{?}{=} \rightarrow$ $\quad p_1 = -\nabla f(x_1)$

$\rightarrow$ calculate $\phi(\alpha_1)$

$\vdots$

x_k

$\|\nabla f(x_k)\| \longrightarrow 0$ $\quad \boxed{\min_{\alpha_1} f(x_1 + \alpha_1 p_1)}$

$\|\nabla f(x_k)\| < \varepsilon$ $\quad$ (ex $\varepsilon = 10^{-5}$ - -)

* If f is quadratic

If $\min_x \underbrace{\frac{1}{2} x^T Q x - b^T x}_{f(x)}$ $\quad$ Q: hessian

$$\boxed{\alpha_k = \frac{\nabla f(x_k)^T \nabla f(x_k)}{\nabla f(x_k)^T Q \nabla f(x_k)}} \longrightarrow \boxed{\text{Exact line search.}}$$

Q: hessian.

$Q = \begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix}$

$\alpha_1 = \dfrac{[4 \;\; 4] \begin{bmatrix} 4 \\ 4 \end{bmatrix}}{\underbrace{\underset{1\times 2}{[4 \;\; 4]} \underset{2\times 2}{\begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix}}}_{1\times 2} \begin{bmatrix} 4 \\ 4 \end{bmatrix}} = \dfrac{32}{[16 \;\; 0] \begin{bmatrix} 4 \\ 4 \end{bmatrix}} = \dfrac{\cancel{32}\,1}{\cancel{64}\,2} = \dfrac{1}{2}$

$\frac{1}{2} x^T Q x - b^T x \overset{?}{=} 4x^2 - 4xy + 2y^2$

$\frac{1}{2}(2x^2 - 2xy + y^2)$

$= \frac{1}{2} [x \;\; y] \begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - b^T \begin{bmatrix} x \\ y \end{bmatrix}$ $\quad$ Hessian. $\quad [0 \;\; 0]$

$\frac{1}{2} \underbrace{[8x - 4y \quad -4x + 4y]}_{1\times 2} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{2\times 1}$

$\frac{1}{2}(8x^2 - 4xy - 4xy + 4y^2)$

$= 4x^2 - 2xy - 2xy + 2y^2$

$= 4x^2 - 4xy + 2y^2$

Ex Use steepest descent method for

$f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1 x_2 + x_2^2$

with $k=0$ $\quad x^0 = (0,0)$ initial point.

$\nabla f(x_1, x_2) = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$ $\quad \nabla f(0,0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\nabla^2 f(x_1, x_2) = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$

$\alpha_0 = \dfrac{[1 \;\; -1] \begin{bmatrix} 1 \\ -1 \end{bmatrix}}{[1 \;\; -1] \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}$

$= \dfrac{2}{[2 \;\; 0] \begin{bmatrix} 1 \\ -1 \end{bmatrix}} = \dfrac{2}{2} = 1$ $\quad p_0 = -\nabla f(0,0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$(k=1)$ 1st iteration

$x^1 = x^0 + \alpha_0 p_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ //

$\nabla f(x^1) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ $\quad \|\nabla f(x^1)\| = \sqrt{2}$ bip. !

move $x^2 = ?$